

MATH1141: HIGHER MATHEMATICS 1A
ASSIGNMENT (PURE MATHEMATICS FLAVOUR)

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QUESTION 1

Let Π_1, Π_2 be a pair of planes in $\mathbb{R}^3$,

$$\begin{aligned}\Pi_1 : \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} &= \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} + \lambda_1 \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} + \mu_1 \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} \\ \Pi_2 : \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} &= \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} + \mu_2 \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix},\end{aligned}$$

where $x_1, x_2, x_3, \lambda_1, \lambda_2, \mu_1, \mu_2 \in \mathbb{R}$. Let L be the line of intersection of Π_1 and Π_2 .

It is given that $\mathbf{n}_1 = \begin{pmatrix} -3 \\ -4 \\ 1 \end{pmatrix}$ and $\mathbf{n}_2 = \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix}$ are vectors normal to Π_1 and Π_2 respectively.

(a) Explain,

(i) how cross products can be used to find $\mathbf{n}_1$ and $\mathbf{n}_2$, and

The cross product of two vectors in $\mathbb{R}^3$ results in a vector orthogonal to both of the vectors (in the direction given by the right hand rule). Hence, the cross product of the direction vectors of each plane yields $\mathbf{n}_1$ and $\mathbf{n}_2$ as follows:

$$\begin{aligned}\mathbf{n}_1 &= \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -4 - (-1) \\ -(6 - 2) \\ -3 - (-4) \end{pmatrix} = \begin{pmatrix} -3 \\ -4 \\ 1 \end{pmatrix}, \quad \text{and} \\ \mathbf{n}_2 &= \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} = \begin{pmatrix} -2 - (-6) \\ -(-4 - (-3)) \\ 4 - 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix}.\end{aligned}$$

(ii) how dot products can be used to check that $\mathbf{n}_1$ and $\mathbf{n}_2$ are indeed normal to their respective planes.

Since $\mathbf{n}_1$ is the normal to the plane Π_1 , it must be orthogonal to both direction vectors of Π_1 , $\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$ and

$\begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$. Similarly, $\mathbf{n}_2$ is orthogonal to both $\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}$.

Consider two non-zero vectors $\mathbf{a} = \overrightarrow{OA}$, $\mathbf{b} = \overrightarrow{OB} \in \mathbb{R}^n$. The cosine rule states

$$|\mathbf{b} - \mathbf{a}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}||\mathbf{b}| \cos \theta.$$

The formula for the length of a vector is

$$\sum_i (b_i - a_i)^2 = \sum_i a_i^2 + \sum_i b_i^2 - 2|\mathbf{a}||\mathbf{b}| \cos \theta.$$

Expanding this provides

$$\sum_i b_i^2 - \sum_i -2a_i b_i + \sum_i a_i^2 = \sum_i a_i^2 + \sum_i b_i^2 - 2|\mathbf{a}||\mathbf{b}| \cos \theta,$$

where terms $\sum_i a_i^2$ and $\sum_i b_i^2$ can be cancelled to produce

$$\sum_i -2a_i b_i = -2|\mathbf{a}||\mathbf{b}| \cos \theta,$$

and finally simplified to

$$\cos \theta = \frac{\sum_i a_i b_i}{|\mathbf{a}||\mathbf{b}|}.$$

The dot product of two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ is defined to be

$$\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i,$$

hence the angle between two non zero vectors $\mathbf{a} = \overrightarrow{OA}$, $\mathbf{b} = \overrightarrow{OB} \in \mathbb{R}^n$ is

$$\theta = \cos^{-1} \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} \right).$$

The vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal when $\theta = \frac{\pi}{2}$, i.e. $\mathbf{a} \cdot \mathbf{b} = 0$

This result can be applied to the vectors $\mathbf{n}_1$ and $\mathbf{n}_2$ by verifying

$$\begin{aligned} \mathbf{n}_1 \cdot \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} &= \mathbf{n}_1 \cdot \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} = 0, \text{ and} \\ \mathbf{n}_2 \cdot \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} &= \mathbf{n}_2 \cdot \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} = 0. \end{aligned}$$

(b) Calculate $\mathbf{m} = \mathbf{n}_1 \times \mathbf{n}_2$.

Using the formula for cross product, we have

$$\mathbf{m} = \begin{pmatrix} -3 \\ -4 \\ 1 \end{pmatrix} \times \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} (-4) \cdot 3 - 1 \cdot 1 \\ 4 \cdot 1 - (-3) \cdot 3 \\ -3 \cdot 1 - 4 \cdot (-4) \end{pmatrix} = \begin{pmatrix} -13 \\ 13 \\ 13 \end{pmatrix}.$$

(c) Using $\mathbf{m}$ as defined in part (b), explain,

(i) how you can tell from $\mathbf{m}$ that the planes are not parallel and hence must intersect in a line, and

Consider two parallel vectors $\begin{pmatrix} x \\ y \\ z \end{pmatrix}, \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \mathbb{R}^3, \lambda \in \mathbb{R}$. Their cross product can be calculated as

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \times \lambda \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} \times \begin{pmatrix} x \\ y \\ z \end{pmatrix} \right) = \mathbf{0}.$$

Therefore, two parallel vectors have a cross product of $\mathbf{0}$.

Since the cross product of the normal vectors $\mathbf{m} = \mathbf{n}_1 \times \mathbf{n}_2 \neq \mathbf{0}$, the two planes are not parallel. Hence, the two planes must intersect in a line.

(ii) why $\mathbf{m}$ must be a vector parallel to the line of intersection of Π_1 and Π_2 .

The line of intersection of the planes Π_1 and Π_2 exists on both planes, hence must be orthogonal to the normal of each plane. The vector $\mathbf{m}$ is the cross product of these normals and therefore must be parallel to the line of intersection of the planes.

(d) You are given that a Cartesian equation for Π_1 is $-3x_1 - 4x_2 + x_3 = 6$. Explain how to use the point-normal form of a plane to find a Cartesian equation for Π_2 , and use this method to find a Cartesian equation for Π_2 .

A plane can be defined using the point normal form as the set of all points $\mathbf{x}$ such that

$$\mathbf{n} \cdot (\mathbf{x} - \mathbf{a}) = 0,$$

where $\mathbf{n}$ is a vector normal to the plane and $\mathbf{a}$ is a point on the plane. The Cartesian equation for such plane

can be obtained by defining $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ and computing the dot product.

Hence, the plane Π_2 can be written as

$$\begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix} \cdot \left(\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} \right) = 0.$$

Computing the dot product yields the Cartesian form of Π_2 ,

$$4x_1 + x_2 - 1 + 3x_3 + 9 = 0,$$

which can be finally simplified to

$$4x_1 + x_2 + 3x_3 = -8.$$

Therefore, the Cartesian equation of the plane Π_2 is $4x_1 + x_2 + 3x_3 = -8$.

- (e) Substitute expressions for x_1 , x_2 and x_3 from the parametric form of Π_2 , and use this method to find a Cartesian equation for Π_1 given in the previous part to find a relationship between λ_1 and μ_2 and use this to find a parametric vector form of the line of intersection L .

The plane Π_2 can be expressed as

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \lambda_2 + 2\mu_2 \\ 1 + 2\lambda_2 + \mu_2 \\ -3 - 2\lambda_2 - 3\mu_2 \end{pmatrix}, \quad \lambda_2, \mu_2 \in \mathbb{R}.$$

Substituting the components of Π_2 into the Cartesian form of Π_1 provides

$$6 = -3(\lambda_2 + 2\mu_2) - 4(1 + 2\lambda_2 + \mu_2) + (-3 - 2\lambda_2 - 3\mu_2),$$

which can be simplified into the relation

$$\lambda_2 + \mu_2 = -1.$$

Substituting $\mu_2 = -\lambda_2 - 1$ into Π_2 provides the parametric equation of the line L

$$\begin{aligned} L : \mathbf{x} &= \begin{pmatrix} 0 \\ 1 \\ -3 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} + (-\lambda_2 - 1) \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix}, \\ &= \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}. \end{aligned}$$

- (f) Explain how you can confirm that your parametric form for the line of intersection is correct by referring to
(i) part (c) to check that the direction is correct, and

The direction of the vector L must be parallel to the vector $\mathbf{m}$ since it is orthogonal to the normal of

both planes. Since the vector L can be written as a scalar multiple of $\mathbf{m}$ as follows:

$$m = \begin{pmatrix} -13 \\ 13 \\ 13 \end{pmatrix} = \mu \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}, \quad \mu = 13,$$

the direction of L is confirmed.

(ii) part (d) to check that one point on your line is correct

The point on the line L must satisfy

$$\Pi_1 : -3x_1 - 4x_2 + x_3 = 6,$$

which can be verified by substituting the found point, $\begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix}$ into Π_1 , providing

$$-3(-2) - 4(0) + 0 = 6,$$

verifying the found point.

QUESTION 2

Show that the equation

$$e^{-3x} + 7 \cos(53x) = 0,$$

has a unique solution for $x \in [0, \frac{\pi}{53}]$.

Let $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^{-3x} + 7 \cos(53x)$. Since $f(x)$ is continuous for $x \in [0, \frac{\pi}{53}]$, the Intermediate Value Theorem (IVT) can be used. IVT states that if a function f is continuous on a closed interval $[a, b]$, it must take every value y between $f(a)$ and $f(b)$ at least once within the interval. Substituting the points 0 and $\frac{\pi}{53}$ into f as follows

$$f(0) = 8 > 0, \text{ and}$$

$$f\left(\frac{\pi}{53}\right) = e^{-\frac{7\pi}{53}} - 7 < 0,$$

yields a positive and negative value respectively, therefore, since $f(0) > 0 > f(\frac{\pi}{53})$, by IVT there is at least one value $c \in (0, \frac{\pi}{53})$ such that

$$f(c) = e^{-3c} + 7 \cos(53c) = 0.$$

Taking the derivative of $f(x)$, we obtain

$$f'(x) = -3e^{-3x} - (7)(53) \sin(53x).$$

This derivative is monotonically decreasing for $x \in (0, \frac{\pi}{53})$ and therefore there exists only one unique value c such that $f(c) = e^{-3c} + 7 \cos(53c) = 0$. Therefore, the equation $e^{-3x} + 7 \cos(53x) = 0$ only has one unique solution for $x \in [0, \frac{\pi}{53}]$.

QUESTION 3

Using the Mean Value Theorem, estimate the error of the approximation of $(243.014)^{\frac{4}{5}}$ to $243^{\frac{4}{5}}$.

The Mean Value Theorem states that for a function f that is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , there exists at least one point c in (a, b) such that $f'(c)$ is equal to the average rate of change over the interval, i.e.

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Let $f : \mathbb{R} \rightarrow \mathbb{R}$, be defined by $f(x) = x^{\frac{4}{5}}$. Since f is an elementary function, it is continuous on $[243, 243.014]$, and happens to be differentiable on $(243, 243.014)$, MVT can be applied.

By MVT, there exists $c \in (243, 243.014)$ such that

$$\frac{f(243.014) - f(243)}{243.014 - 243} = f'(c) = \frac{4}{5}c^{-\frac{1}{5}},$$

which can be simplified to

$$(243.014)^{\frac{4}{5}} - 243^{\frac{4}{5}} = \frac{4}{5}c^{-\frac{1}{5}}(243.014 - 243).$$

Since $f(x)$ is monotonically increasing between $[243, 243.014]$, the error in approximation is highest at $c = 243$. Hence the approximation error is given by

$$\frac{4}{5} \cdot 243^{-\frac{1}{5}} \cdot (243.014 - 243) = \frac{4}{15} \cdot 0.014 = \frac{7}{1875}.$$

Therefore, the magnitude of error in making the approximation of $(243.014)^{\frac{4}{5}}$ as $243^{\frac{4}{5}}$ is at most $\frac{7}{1875}$.